

Compute the summations and products in 19-28.

5.1.22. $\prod_{j=0}^4 (-1)^j$

$$\prod_{j=0}^4 (-1)^j = (-1)^0 (-1)^1 (-1)^2 (-1)^3 (-1)^4$$

$$(-1)^0 = 1$$

$$(-1)^1 = (-1)$$

$$(-1)^2 = 1$$

$$(-1)^3 = (-1)$$

$$(-1)^4 = 1$$

$$1 * (-1) * 1 * (-1) * 1 = 1$$

Solution: $\prod_{j=0}^4 (-1)^j = 1$